1.1 Notes — Proofs, logic and sets

These notes condense the module. They are not a replacement for the exercises: proof-writing is learned by writing proofs. Read a section, close the notes, and try to reproduce the key proofs.

Convention: $\mathbb{N} = \{1, 2, 3, \dots\}$. Where 0 is needed we write $\mathbb{N}_0$.

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1. Logic

1.1 Connectives

A *statement* is a sentence that is either true or false. From statements P, Q we form $\neg P$ (not), $P \wedge Q$ (and), $P \vee Q$ (inclusive or), $P \Rightarrow Q$ (if… then) and $P \Leftrightarrow Q$ (if and only if).

The only connective whose truth table people get wrong is the implication:

P	Q	$P \Rightarrow Q$
T	T	T
T	F	F
F	T	T
F	F	T

An implication with a false hypothesis is *vacuously true*. This is why "every element of $\emptyset$ is purple" is true, and why $\emptyset$ is a subset of every set.

Key equivalences (check each with a truth table once, then memorise):

$$P \Rightarrow Q \;\equiv\; \neg P \vee Q \;\equiv\; \neg Q \Rightarrow \neg P \quad \text{(contrapositive)}$$

$$\neg(P \wedge Q) \equiv \neg P \vee \neg Q, \qquad \neg(P \vee Q) \equiv \neg P \wedge \neg Q \quad \text{(De Morgan)}$$

$$\neg(P \Rightarrow Q) \equiv P \wedge \neg Q.$$

The **converse** $Q \Rightarrow P$ is *not* equivalent to $P \Rightarrow Q$. Confusing an implication with its converse is the most common logical error in proofs.

1.2 Quantifiers

$\forall x \in S : P(x)$ means "for every x in S, $P(x)$". $\exists x \in S : P(x)$ means "there is at least one x in S with $P(x)$". $\exists!$ means "there is exactly one".

Negation rules:

$$\neg\left(\forall x : P(x)\right) \equiv \exists x : \neg P(x), \qquad \neg\left(\exists x : P(x)\right) \equiv \forall x : \neg P(x).$$

For nested quantifiers, apply the rule from the outside in: each quantifier flips and the negation moves inwards. Bounded quantifiers keep their bounds. The negation of $\forall \varepsilon > 0 \ \dots$ is $\exists \varepsilon > 0 \ \dots$, *not* $\exists \varepsilon \leq 0$.

Example (Cauchy sequence). (a_n) is Cauchy if

$$\forall \varepsilon > 0 \; \exists N \in \mathbb{N} \; \forall m, n \geq N : |a_m - a_n| < \varepsilon.$$

It is *not* Cauchy if

$$\exists \varepsilon > 0 \; \forall N \in \mathbb{N} \; \exists m, n \geq N : |a_m - a_n| \geq \varepsilon.$$

Order matters. $\forall x \in \mathbb{R} \; \exists y \in \mathbb{R} : y > x$ is true: y may depend on x. $\exists y \in \mathbb{R} \; \forall x \in \mathbb{R} : y > x$ is false: a single y would have to beat everything, including y itself. In general $\exists y \forall x$ implies $\forall x \exists y$, but not conversely. The difference between continuity and *uniform* continuity (module 1.2) is exactly this swap.

1.3 Reading the logical form of a goal

The logical form of what you want to prove dictates the opening of the proof (Velleman's central idea):

Goal	Proof starts with
$P \Rightarrow Q$	"Assume P." Then derive Q.
$\forall x \in S : P(x)$	"Let $x \in S$ be arbitrary." Then prove $P(x)$.
$\exists x : P(x)$	Construct or exhibit an x and verify $P(x)$, or prove existence indirectly.
$P \wedge Q$	Prove P; prove Q.
$P \vee Q$	"Assume $\neg P$." Then prove Q. Or split into cases.
$P \Leftrightarrow Q$	Prove both directions.
$A = B$ (sets)	Prove $A \subseteq B$ and $B \subseteq A$.

Hypotheses are *used* in the dual way. From $\forall x : P(x)$ you may plug in any x you like. From $\exists x : P(x)$ you may say "choose such an x", and it is then fixed and you do not control it.

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2. Proof techniques

2.1 Direct proof

Assume the hypothesis and chain implications to the conclusion.

Example. If $a \mid b$ and $b \mid c$ then $a \mid c$. *Proof:* $b = ak$ and $c = bl$ for integers k, l, so $c = a(kl)$. $\square$

2.2 Contrapositive

To prove $P \Rightarrow Q$, prove $\neg Q \Rightarrow \neg P$. Use it when $\neg Q$ gives you something concrete to work with.

Example. If n^2 is even then n is even. *Proof:* suppose n is odd, $n = 2k + 1$. Then $n^2 = 2(2k^2 + 2k) + 1$ is odd. $\square$

2.3 Contradiction

Assume the statement is false and derive an absurdity.

Theorem. $\sqrt{2}$ is irrational.

Proof. Suppose $\sqrt{2} = p/q$ with p, q positive integers and the fraction in lowest terms. Then $p^2 = 2q^2$, so p^2 is even, so p is even (by §2.2). Write $p = 2r$. Then $4r^2 = 2q^2$, so $q^2 = 2r^2$ and q is even too, contradicting lowest terms. □

Theorem (Euclid). There are infinitely many primes.

Proof. Suppose $p_1, \dots, p_k$ are all the primes, and let $N = p_1 p_2 \cdots p_k + 1$. Since $N > 1$, it has a prime factor p, which must be some p_i. Then p_i divides both N and $N - 1$, so $p_i \mid 1$, which is absurd. □

Style note. Many "proofs by contradiction" are really contrapositive proofs in disguise: you assume P and $\neg Q$, derive $\neg P$, and declare a contradiction. Rewrite these as contrapositives; it is cleaner. Keep contradiction for genuinely indirect arguments, such as irrationality, infinitude, or uncountability.

2.4 Cases

Split into exhaustive cases and prove each. *Example:* $n^2 + n$ is even for every integer n (case n even, case n odd). Better: $n^2 + n = n(n + 1)$ is a product of consecutive integers, one of which is even. Look for the argument that avoids cases.

2.5 Existence and uniqueness

To prove $\exists! x : P(x)$, prove existence (find one) and uniqueness (if $P(x)$ and $P(y)$ then $x = y$).

A **non-constructive** existence proof shows something exists without exhibiting it. *Classic:* there are irrationals a, b with a^b rational. Either $\sqrt{2}^{\sqrt{2}}$ is rational (take $a = b = \sqrt{2}$), or it is not, and then take $a = \sqrt{2}^{\sqrt{2}}$, $b = \sqrt{2}$, so that $a^b = \sqrt{2}^2 = 2$. We do not learn which case holds.

2.6 Disproof

To disprove $\forall x : P(x)$, give *one* x with $\neg P(x)$. To disprove $\exists x : P(x)$, prove $\forall x : \neg P(x)$, which needs a real proof.

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3. Induction

3.1 Ordinary induction

Principle. If $P(1)$ holds and $P(n) \Rightarrow P(n + 1)$ for every $n \in \mathbb{N}$, then $P(n)$ holds for all $n \in \mathbb{N}$.

Template. "We prove $P(n)$ by induction on n. **Base case:** ... **Inductive step:** let $n \geq 1$ and assume $P(n)$ (the inductive hypothesis). Then ... [using the hypothesis] ... so $P(n + 1)$ holds."

Example. For $x > -1$ and all $n \in \mathbb{N}$, $(1 + x)^n \geq 1 + nx$ (Bernoulli's inequality). Inductive step: $(1 + x)^{n+1} = (1 + x)^n (1 + x) \geq (1 + nx)(1 + x) = 1 + (n + 1)x + nx^2 \geq 1 + (n + 1)x$. The first inequality uses $1 + x > 0$.

3.2 Strong induction

Principle. If, for every n, the truth of $P(k)$ for all $k < n$ implies $P(n)$, then $P(n)$ holds for all n.

Example (Fundamental Theorem of Arithmetic, existence). Every $n \geq 2$ is a product of primes. If n is prime, done. Otherwise $n = ab$ with $2 \leq a, b < n$, and by the strong hypothesis both a and b are products of primes, hence so is n.

3.3 Well-ordering

Well-ordering principle. Every non-empty subset of $\mathbb{N}$ has a least element.

Ordinary induction, strong induction and well-ordering are equivalent (each can be proved from the others). Well-ordering gives proof by **smallest counterexample**: if $P(n)$ fails for some n, take the least such n and derive a contradiction by finding a smaller one.

3.4 Pitfalls

- Forgetting or fudging the base case. ("All horses are the same colour" breaks at $n = 1 \to 2$: the inductive step silently assumes two overlapping groups.)
- Never using the inductive hypothesis. If you did not use it, you did not need induction, or you have made an error.
- Inducting on the wrong variable, or proving $P(n+1) \Rightarrow P(n)$ by accident.

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4. Sets

4.1 Basics

A set is determined by its elements (extensionality). $A \subseteq B$ means $\forall x : x \in A \Rightarrow x \in B$. To prove $A = B$, prove $A \subseteq B$ and $B \subseteq A$ (*double inclusion*, or element-chasing).

Operations: $A \cup B$, $A \cap B$, $A \setminus B$, the complement A^c (relative to a universe U), the power set $\mathcal{P}(A)$, and the Cartesian product $A \times B = \{(a, b) : a \in A, b \in B\}$.

De Morgan's laws for an indexed family $(A_i)_{i \in I}$:

$$\left(\bigcup_{i \in I} A_i\right)^c = \bigcap_{i \in I} A_i^c, \qquad \left(\bigcap_{i \in I} A_i\right)^c = \bigcup_{i \in I} A_i^c.$$

Proof of the first: $x \notin \bigcup A_i \iff \neg(\exists i : x \in A_i) \iff \forall i : x \notin A_i \iff x \in \bigcap A_i^c$. Set identities are logical identities in disguise.

Indexed families are used constantly in analysis and measure theory. For example, $\bigcap_{n \geq 1} (0, 1/n) = \emptyset$ and $\bigcup_{n \geq 1} [1/n, 1] = (0, 1]$. Check both carefully: these are where intuition about "the limit" misleads.

4.2 Russell's paradox

The "set" $R = \{x : x \notin x\}$ would satisfy $R \in R \iff R \notin R$. The fix (Zermelo–Fraenkel set theory, ZF) is that you may only form subsets $\{x \in A : P(x)\}$ of a set A you already have. In practice: always know which set your elements are drawn from.

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5. Functions

5.1 Definitions

A function $f : A \to B$ is a subset $f \subseteq A \times B$ such that every $a \in A$ is paired with exactly one $b \in B$. A is the domain and B the codomain. The codomain is part of the data: $x \mapsto x^2$ is surjective onto $[0, \infty)$ but not onto $\mathbb{R}$.

- **Injective:** $f(a) = f(a') \Rightarrow a = a'$.
- **Surjective:** $\forall b \in B\ \exists a \in A : f(a) = b$.
- **Bijective:** both. This holds exactly when there is an inverse $f^{-1} : B \to A$ with $f^{-1} \circ f = \mathrm{id}_A$ and $f \circ f^{-1} = \mathrm{id}_B$.

Composition facts (prove each once):

1. If $g \circ f$ is injective, then f is injective.
2. If $g \circ f$ is surjective, then g is surjective.
3. A composition of injections is injective, and a composition of surjections is surjective.
4. f has a left inverse ($h \circ f = \mathrm{id}$) iff f is injective (for $A \neq \emptyset$). f has a right inverse ($f \circ h = \mathrm{id}$) iff f is surjective. The second needs the axiom of choice in general: you must *choose* a preimage for each b.

5.2 Images and preimages

For $X \subseteq A$ and $Y \subseteq B$: $f(X) = \{f(x) : x \in X\}$ and $f^{-1}(Y) = \{a \in A : f(a) \in Y\}$. The preimage is defined for *every* f, not only bijections.

Preimages preserve everything:

$$f^{-1}\left(\bigcup Y_i\right) = \bigcup f^{-1}(Y_i), \quad f^{-1}\left(\bigcap Y_i\right) = \bigcap f^{-1}(Y_i), \quad f^{-1}(B \setminus Y) = A \setminus f^{-1}(Y).$$

Images preserve unions only: $f(\bigcup X_i) = \bigcup f(X_i)$, but in general only $f(X_1 \cap X_2) \subseteq f(X_1) \cap f(X_2)$, with equality for all X_1, X_2 iff f is injective.

Round trips:

$f\left(f^{-1}(Y)\right) \subseteq Y$ (equality for all $Y \iff f$ surjective), $\qquad X \subseteq f^{-1}\left(f(X)\right)$ (equality for all $X \iff f$ inject

These facts explain why topology and measure theory define continuity and measurability using **preimages**: preimages respect all the set operations used to build open sets and σ-algebras.

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6. Relations

A relation on A is a subset $R \subseteq A \times A$; write $a \sim b$ for $(a, b) \in R$.

Property	Meaning
Reflexive	$a \sim a$ for all a
Symmetric	$a \sim b \Rightarrow b \sim a$
Antisymmetric	$a \sim b$ and $b \sim a \Rightarrow a = b$
Transitive	$a \sim b$ and $b \sim c \Rightarrow a \sim c$

6.1 Equivalence relations

An **equivalence relation** is reflexive, symmetric and transitive. The **class** of a is $[a] = \{x \in A : x \sim a\}$, and the quotient set is $A/\sim = \{[a] : a \in A\}$.

Theorem. The equivalence classes of an equivalence relation partition A: they are non-empty, cover A, and any two are either equal or disjoint. Conversely, every partition of A comes from exactly one equivalence relation ("lie in the same block").

Proof sketch. $a \in [a]$ by reflexivity, so the classes are non-empty and cover A. If $c \in [a] \cap [b]$, then for $x \in [a]$ we get $x \sim a \sim c \sim b$ (using symmetry for $a \sim c$), so $x \in [b]$. Hence $[a] \subseteq [b]$, and by symmetry $[a] = [b]$. □

Key examples:

- **Congruence mod** n on $\mathbb{Z}$: $a \equiv b$ iff $n \mid (a - b)$. The quotient is $\mathbb{Z}/n\mathbb{Z} = \{[0], \dots, [n-1]\}$.
- **Construction of** $\mathbb{Q}$**:** on $\mathbb{Z} \times (\mathbb{Z} \setminus \{0\})$, $(a, b) \sim (c, d)$ iff $ad = bc$. The class of (a, b) "is" a/b.
- **Projective space:** non-zero vectors of $\mathbb{R}^{n+1}$ modulo scaling by non-zero reals.
- **Cosets** gH of a subgroup (module 1.4). **Quotient spaces** V/W (module 1.3). **Functions equal almost everywhere** (module 3.1, where L^p is a quotient).

6.2 Well-definedness

When you define a function on a quotient by a formula in a representative, $F([a]) := \text{formula}(a)$, you must check that the result does not depend on the representative: $a \sim a' \Rightarrow \text{formula}(a) = \text{formula}(a')$.

Example. On $\mathbb{Z}/n\mathbb{Z}$, $[a] + [b] := [a + b]$ is well defined: if $a' = a + kn$ and $b' = b + ln$ then $a' + b' = a + b + (k + l)n$. By contrast, $F([a]) := a$ is *not* well defined: it depends on the representative.

This check appears throughout algebra and is one of the most often omitted steps.

6.3 Orders

A **partial order** is reflexive, antisymmetric and transitive; a **total order** also has every two elements comparable. Examples: $\leq$ on $\mathbb{R}$ (total); $\subseteq$ on $\mathcal{P}(A)$ and divisibility on $\mathbb{N}$ (partial, not total).

In a poset $(P, \leq)$:

- An **upper bound** of $S \subseteq P$ is some $u \in P$ with $s \leq u$ for all $s \in S$.
- A **greatest** element of P is above everything. A **maximal** element has nothing strictly above it.
- A **chain** is a totally ordered subset.

Greatest implies maximal, but not conversely: in $\{2, 3, \dots, 10\}$ under divisibility, $6, 7, 8, 9, 10$ are all maximal and there is no greatest element.

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7. Cardinality

7.1 Definitions

$|A| = |B|$ means there is a bijection $A \to B$. $|A| \leq |B|$ means there is an injection $A \to B$. A is **finite** if it is empty or in bijection with $\{1, \dots, n\}$ for some n, **countably infinite** if $|A| = |\mathbb{N}|$, **countable** if finite or countably infinite, and **uncountable** otherwise.

Useful facts:

- A subset of a countable set is countable.
- A is countable iff there is an injection $A \to \mathbb{N}$, iff ($A = \emptyset$ or there is a surjection $\mathbb{N} \to A$).
- An infinite set has a countably infinite subset (this uses a weak form of choice). So $\aleph_0 = |\mathbb{N}|$ is the smallest infinite cardinal.

7.2 Countable sets

$\mathbb{N} \times \mathbb{N}$ **is countable.** $(m, n) \mapsto 2^m 3^n$ is injective by unique factorisation. Alternatively, list the pairs along anti-diagonals $m + n = 2, 3, 4, ...$, which gives the explicit bijection $(m, n) \mapsto \binom{m+n-1}{2} + m$.

$\mathbb{Z}$ **and** $\mathbb{Q}$ **are countable.** $\mathbb{Z}$: list $0, 1, -1, 2, -2, ...$ $\mathbb{Q}$: the map $\mathbb{Z} \times \mathbb{N} \to \mathbb{Q}$, $(p, q) \mapsto p/q$, is a surjection from a countable set.

A countable union of countable sets is countable. If each A_n is countable, choose surjections $f_n : \mathbb{N} \to A_n$ (assume each A_n is non-empty). Then $(n, k) \mapsto f_n(k)$ is a surjection $\mathbb{N} \times \mathbb{N} \to \bigcup A_n$. Note that we *chose* infinitely many surjections; this uses the axiom of countable choice. Without some choice this theorem can fail.

Algebraic numbers are countable. There are countably many integer polynomials (a countable union over degree and coefficients), each with finitely many roots. Hence transcendental numbers exist, and in fact almost all reals are transcendental.

7.3 Uncountable sets

Theorem (Cantor, 1891 diagonal argument). $(0, 1)$ is uncountable.

Proof. Let $f : \mathbb{N} \to (0, 1)$ be any function and write $f(n) = 0.d_{n1}d_{n2}d_{n3}...$ in decimal, using the expansion that does not end in recurring 9s. Define $x = 0.e_1e_2e_3...$ with $e_n = 5$ if $d_{nn} \neq 5$ and $e_n = 4$ if $d_{nn} = 5$. Then $x \in (0, 1)$, its expansion is the non-9s one, and x differs from $f(n)$ in the nth digit. So $x \notin f(\mathbb{N})$, and f is not surjective. □

(Using digits 4 and 5 avoids the $0.0999... = 0.1$ ambiguity.)

Theorem (Cantor). For every set A, there is no surjection $A \to \mathcal{P}(A)$. Hence $|A| < |\mathcal{P}(A)|$.

Proof. Given $f : A \to \mathcal{P}(A)$, let $D = \{a \in A : a \notin f(a)\}$. If $D = f(d)$, then $d \in D \iff d \notin f(d) = D$, a contradiction. The map $a \mapsto \{a\}$ is an injection, so $|A| \leq |\mathcal{P}(A)|$, and the inequality is strict. □

So there is an infinite hierarchy $|\mathbb{N}| < |\mathcal{P}(\mathbb{N})| < |\mathcal{P}(\mathcal{P}(\mathbb{N}))| < \cdots$.

7.4 Schröder–Bernstein

Theorem (Schröder–Bernstein). If there are injections $f : A \to B$ and $g : B \to A$, then there is a bijection $h : A \to B$.

So $\leq$ on cardinalities is antisymmetric. The theorem does not use the axiom of choice.

Proof. Let $C_0 = A \setminus g(B)$ (the points of A not hit by g), $C_{n+1} = g(f(C_n))$, and $C = \bigcup_{n \geq 0} C_n$. Define

$$h(a) = \begin{cases} f(a) & a \in C, \\ g^{-1}(a) & a \notin C. \end{cases}$$

This makes sense: if $a \notin C$ then $a \notin C_0$, so $a \in g(B)$, and $g^{-1}(a)$ is the unique preimage because g is injective.

Injective. Each piece is injective. Suppose $a \in C$, $a' \notin C$ and $f(a) = g^{-1}(a')$. Then $a' = g(f(a))$. If $a \in C_n$ then $a' \in C_{n+1} \subseteq C$, a contradiction.

Surjective. Let $b \in B$. If $g(b) \notin C$, then $h(g(b)) = g^{-1}(g(b)) = b$. If $g(b) \in C$, then $g(b) \notin C_0$ (it lies in $g(B)$), so $g(b) \in C_{n+1} = g(f(C_n))$ for some n. Thus $g(b) = g(f(a))$ with $a \in C_n \subseteq C$. Since g is injective, $b = f(a) = h(a)$. □

Picture. Follow each point backwards along the chain $\cdots \to a \xrightarrow{f} b \xrightarrow{g} a' \to \cdots$. Chains that start in $A \setminus g(B)$ are matched using f; all others are matched using g^{-1}.

Applications.

- $|[0,1]| = |(0,1)| = |\mathbb{R}|$. Injections are easy to write down; bijections are fiddly.
- $|\mathbb{R}| = |\mathcal{P}(\mathbb{N})|$. One injection: $\mathcal{P}(\mathbb{N}) \to \mathbb{R}$, $S \mapsto \sum_{n \in S} 10^{-n}$ (decimal digits 0 and 1 only, so no two sets give the same number). The other: $\mathbb{R} \to \mathcal{P}(\mathbb{Q})$, $x \mapsto \{q \in \mathbb{Q} : q < x\}$ (a Dedekind cut), together with $|\mathcal{P}(\mathbb{Q})| = |\mathcal{P}(\mathbb{N})|$.

The **continuum hypothesis** (no cardinality strictly between $|\mathbb{N}|$ and $|\mathbb{R}|$) can be neither proved nor disproved in ZFC (Gödel 1940, Cohen 1963). You will not need it.

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8. The axiom of choice and Zorn's lemma

Axiom of choice (AC). For any family $(A_i)_{i \in I}$ of non-empty sets, there is a function choosing one element from each: $c(i) \in A_i$. It is needed only when the family is infinite *and* there is no rule to make the choice.

Zorn's lemma. Let $(P, \leq)$ be a non-empty partially ordered set in which every chain has an upper bound in P. Then P has a maximal element.

Zorn's lemma is equivalent to AC (and to the well-ordering theorem), given the other ZF axioms. You are not expected to prove the equivalence. Learn the *pattern of use*:

1. Let P be the set of "partial objects" (independent sets, proper ideals, partial extensions…), ordered by inclusion.
2. Check that P is non-empty.
3. Check that every chain has an upper bound, usually its union, and verify that the union is still a partial object.
4. Get a maximal element, and show that maximal implies "complete", because otherwise you could extend it.

Theorem. Every vector space V has a basis.

Proof. Let P be the set of linearly independent subsets of V, ordered by $\subseteq$. $\emptyset \in P$, so P is non-empty. If $\mathcal{C}$ is a chain in P, let $U = \bigcup \mathcal{C}$. Any finite subset $\{u_1, \dots, u_k\} \subseteq U$ has each u_j in some $L_j \in \mathcal{C}$. Since $\mathcal{C}$ is a chain, one of the L_j contains all the others, so $u_1, \dots, u_k$ lie in a single independent set and are independent. Hence $U \in P$ is an upper bound. By Zorn's lemma there is a maximal independent set B. If some $v \notin \operatorname{span} B$, then $B \cup \{v\}$ would be independent, contradicting maximality. So B spans V and is a basis. □

Other places Zorn's lemma will appear: every proper ideal of a ring with 1 lies in a maximal ideal (3.3), the Hahn–Banach theorem (3.2), and Tychonoff's theorem (2.1 and beyond). It is also used to show there exist non-measurable sets (3.1), via a choice of representatives for $\mathbb{R}/\mathbb{Q}$ (the Vitali set).

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9. Common pitfalls

1. **Proving the converse.** You were asked for $P \Rightarrow Q$ and proved $Q \Rightarrow P$. Write the goal at the top of the page before starting.
2. **"Let ε be small."** In "for all ε" proofs, ε is arbitrary and fixed. You do not get to choose it.
3. **Dependence of choices.** In $\forall x\, \exists y$, the y may depend on x; make the dependence explicit ($N = N(\varepsilon)$, $\delta = \delta(\varepsilon, x)$).
4. **Using what you want to prove.** Starting a proof of $A = B$ with "$A = B$, so ..." is circular.
5. **Well-definedness** on quotients (§6.2), and **codomain** awareness for surjectivity (§5.1).
6. **Writing $f^{-1}(Y)$ as if f were invertible.** The preimage always exists; the inverse function may not.
7. **Infinite operations by analogy with finite ones.** $\bigcap_n (0, 1/n) = \emptyset$, not $\{0\}$. Check each element.
8. **Choosing infinitely many things silently.** It is fine to use AC, but know when you are doing so.

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10. Connections forward

Topic here	Where it returns
Quantifier negation	Every ε–δ argument in 1.2, 2.1 and 3.1
Preimages preserve set operations	Definition of continuity in topology (2.1); measurable functions and σ-algebras (3.1)
Equivalence relations and quotients	Cosets and quotient groups (1.4); quotient spaces (1.3); quotient topology and identification spaces (3.4); L^p spaces as functions modulo "equal almost everywhere" (3.1)
Countable versus uncountable	Countable additivity and null sets (3.1); separability of Banach and Hilbert spaces (3.2); the Baire category theorem (2.1, 3.2)
Cantor's diagonal argument	Diagonal subsequence arguments in Arzelà–Ascoli (2.1) and Banach–Alaoglu (3.2)
Zorn's lemma	Maximal ideals (3.3); Hahn–Banach (3.2); bases of infinite-dimensional spaces; non-measurable sets (3.1)
Induction and well-ordering	Euclid's algorithm and unique factorisation (1.4); structural induction in algebraic topology (3.4)